

Hyeonseop Yoon

Applied NLP / AI Researcher · Grounded Retrieval, Agent Systems, Evaluation

Seoul, Korea · xianxie31@korea.ac.kr · pfsv.github.io · GitHub · ORCID

PROFILE

AI research engineer building grounded, evidence-driven language systems under production constraints. Experience spans retrieval and reranking, query expansion, RAG and agent systems, LLM adaptation, evaluation, and on-premise serving. Earlier research connected NLP representations with human cognition and neural signals.

EDUCATION

Hankuk University of Foreign Studies Mar 2019 — Aug 2024
B.A. in English Linguistics & Language Technology · Double major in AI Software Engineering
GPA: 4.17/4.5 · Undergraduate thesis: Metaphor in Mind and Machine

PROFESSIONAL EXPERIENCE

MAUM.AI Mar 2024 — Present
AI Research Engineer · AICC R&D;

- Lead research and engineering across retrieval, RAG, agent orchestration, model adaptation, evaluation, and serving for automotive, insurance, finance, and public-sector systems.
- Developed a cascaded reranking and domain-adaptation pipeline that improved documented FAQ accuracy from 69.4% to 96.3% on a client evaluation set.
- Built verified-unit QA and staged linguistic seeding for two contact-center domains; the resulting application study was accepted to GroundLM at EMNLP 2026.
- Built structure-aware retrieval for constrained policy and technical documents, improving an internal exact-match evaluation from 4/59 to 47/59.
- Delivered on-premise LLM and RAG systems, including model fine-tuning, quantization, synthetic-data generation, offline deployment, distributed serving, and observability.

Seoul National University Dec 2023 — Feb 2024
Researcher · Cognitive & Systems Neuroscience Laboratory

- Reviewed and analyzed RNN models of visual working memory, sensory encoding, and decision bias; inspected training dynamics and visualized unit-level information flow.

Korea University Aug 2022 — Sep 2023
Researcher · Brain Signal Processing Laboratory

- Fine-tuned transformer and recurrent language models for reasoning tasks and extracted hidden representations and attention patterns.
- Compared model representations with voxel-level fMRI signals using representational similarity analysis and whole-brain mapping.

SELECTED PROJECTS

Production Grounded AI Systems 2024 — 2026
MAUM.AI · Automotive, insurance, rail, and public-sector deployments

- K-company automotive AICC: domain-specific FAQ retrieval, slot-filling agent, long-term session memory, model compression, and low-latency inference across multiple service locations.
- K-company rail service: on-premise voice chatbot and offline regulation-search agents using synthetic corpus generation and grounded retrieval.
- K-company insurance systems: document-layout parsing, policy-code mapping agents, knowledge distillation, retrieval over internal regulations, and distributed model serving.

pycag 2026
Open-source Cache-Augmented Generation toolkit
PyPI package and public source: github.com/PFSV/cag
Korean Retrieval Embeddings 2025 — 2026
BGE-M3 and Qwen3-based public encoders
Model cards document training scope, AutoRAG evaluation, and limitations.
VisionCardio 2025
On-device rPPG research prototype
PyTorch training, Core ML export, SwiftUI integration, and deterministic wellness coaching policy.

PUBLICATIONS

[P4] Staged Linguistic Seeding: Grounded Query Expansion for Verified-Unit QA in AI Contact Centers. Hyeonseop Yoon, Jeong-Eun Park. GroundLM Workshop at EMNLP 2026. arXiv:2609.00844

[P3] Metaphor in Mind and Machine. Hyeonseop Yoon et al. OHBM 2024 / Aperture Neuro. DOI

[P2] Comparative Analysis of Brain and NLP Models for Reasoning Tasks. Hyeonseop Yoon et al. Brain Engineering Society of Korea, 2023.

[P1] Korean Twitter Bot Detection based on Deep Learning. Hyeonseop Yoon et al. Korea Software Congress, KIISE, 2022.

ACADEMIC SERVICE

Reviewer	2026
Grounding Language Models (GroundLM) Workshop at EMNLP	
Reviewer	2026
Vision-Language Models for Real-World Deployment (VLM4RWD) Workshop at NeurIPS	

SELECTED HONORS

Outstanding Poster Award	2023
Korean Society for Human Brain Mapping	
Metaphor in Mind and Machine	
Outstanding Poster Award	2023
Brain Engineering Society of Korea	
Comparative Analysis of Brain and NLP Models for Reasoning Tasks	
Outstanding Research Award	2022
Korea Software Congress, KIISE	
Korean Twitter Bot Detection based on Deep Learning	

ADDITIONAL EXPERIENCE & SKILLS

Open Source Contribution Academy Masters (Aug-Dec 2022) — Developed and maintained VisualPython, a Python-based GUI framework for statistics, machine learning, and deep learning workflows.

Data Youth Campus (Jul-Sep 2022) — Completed an NLP and deep-learning convergence program; built a Korean spam-detection system.

Technical: Python, PyTorch, Hugging Face, vLLM, RAG, BM25/SPLADE/dense retrieval, LoRA/ORPO, Docker, PostgreSQL/pgvector, Ray, SLURM/H100, Core ML, SQL, NumPy, pandas, scikit-learn

Language: Korean (native), English (TEPS 394, Feb 2024; lived in the United Kingdom, 2011-2013)

Military status: Not completed